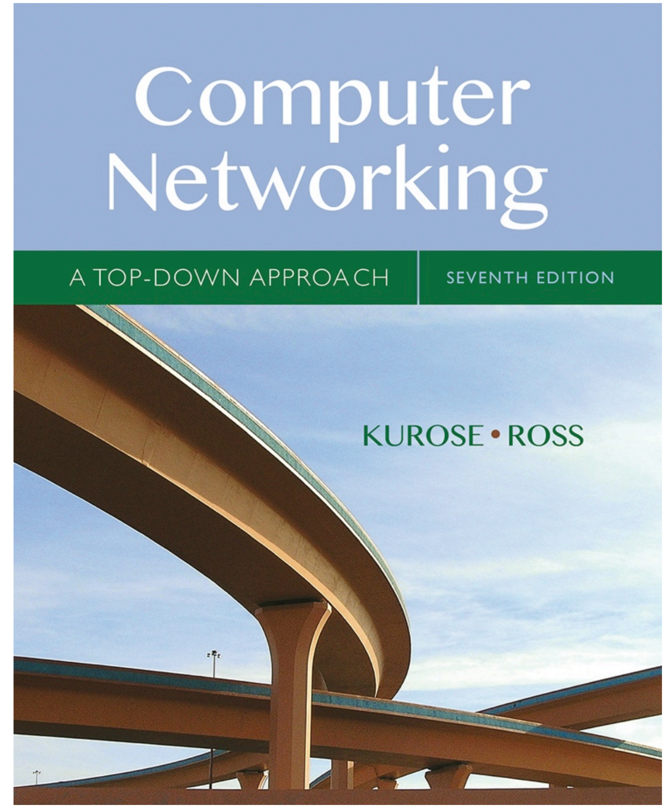
Chapter 5
Network Layer: The Control Plane



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Computer Networking: A Top Down Approach

7th edition
Jim Kurose, Keith Ross
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Chapter 5: network layer control plane

understand principles behind network control plane

- traditional routing algorithms

and their instantiation, implementation in the Internet:

- OSPF, BGP

Chapter 5: outline

5.1 introduction

- Control Plane
- Autonomous Systems

5.2 routing protocols

- link state
- distance vector

5.3 intra-AS routing in the Internet:

- RIP
- OSPF

5.4 inter-AS routing in the Internet: B

Network-layer functions

Recall: two network-layer functions:

- *forwarding:* move packets from router's input to appropriate router output

data plane

- *routing:* determine route taken by packets from source to destination

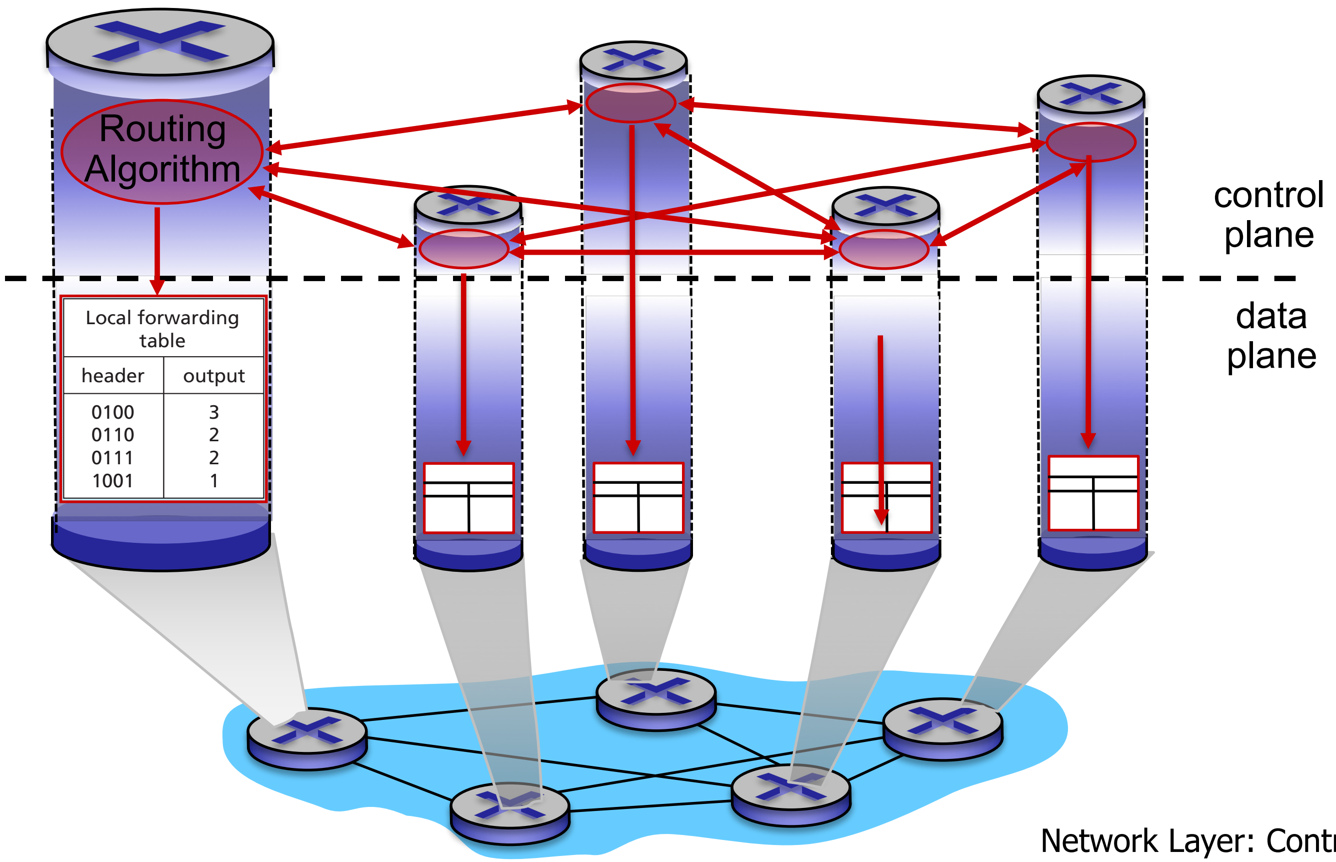
control plane

Two approaches to structuring network control plane:

- per-router control (traditional)
- logically centralized control (software defined networking)

Per-router control plane

Individual routing algorithm components *in each and every router* interact with each other in control plane to compute forwarding tables



Making routing scalable

our routing study thus far - idealized
- all routers identical
- network "flat"

… *not* true in practice

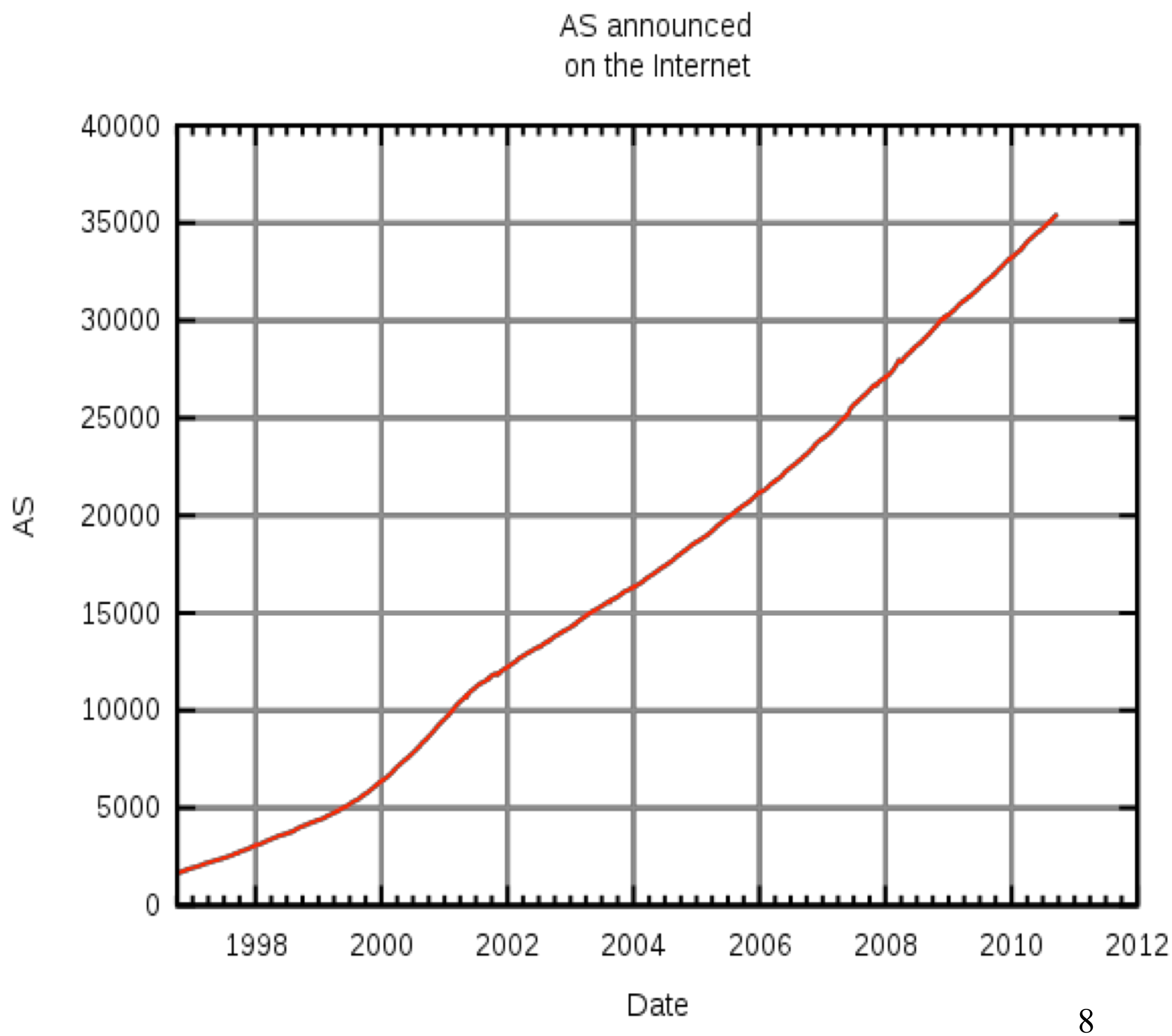
scale: with billions of destinations:
- can't store all destinations in routing tables!
- routing table exchange would swamp links!

administrative autonomy
- internet = network of networks
- each network admin may want to control routing in its own network

Autonomous Systems

- an **autonomous system (AS)** is a region of the Internet that is administered by a single entity and that has a unified routing policy
- each autonomous system is assigned an Autonomous System Number (**ASN**). Each ASN is either 16bits or 32bits
  - ASN assigned by Regional Internet Registries
  - some are reserved for private use and never appear on the Internet
  - example ASNs
    - U of T's campus network (AS239)
    - Sprint (AS1239, AS1240, AS 6211, …)

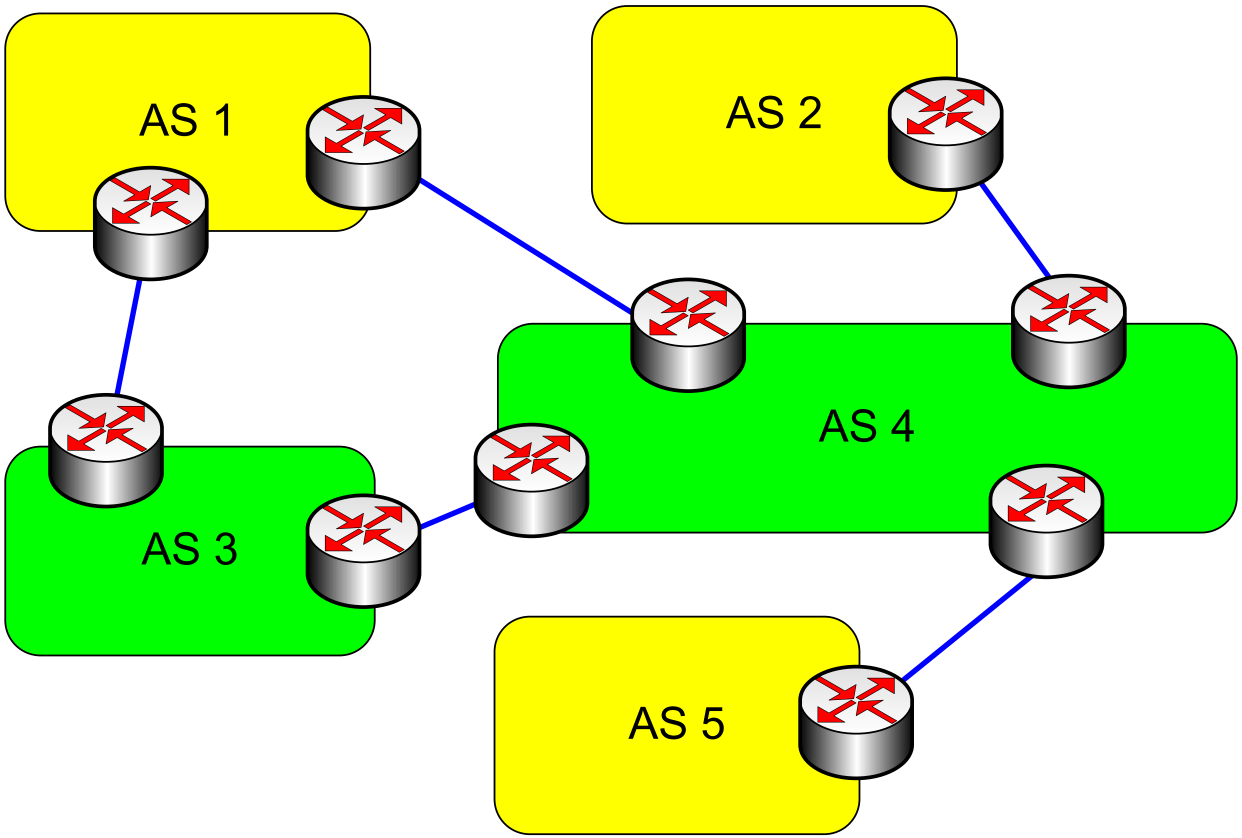
Number of Autonomous Systems



Autonomous Systems terminology

- **local traffic**: traffic with source and destination in AS
- **transit traffic**: traffic that passes through the AS
- **Stub AS**: has connection to only one AS, only carries local traffic
- **Multihomed Stub AS**: has connection to >1 AS, but does not carry transit traffic
- **Transit AS**: has connection to >1 AS and carries transit traffic

Stub and Transit Networks



- AS 1 is a multi-homed stub network
- AS 3 and AS 4 are transit networks
- AS 2 and AS 5 are stub networks

Internet approach to scalable routing

aggregate routers into regions known as "autonomous systems" (AS) (a.k.a. "domains")

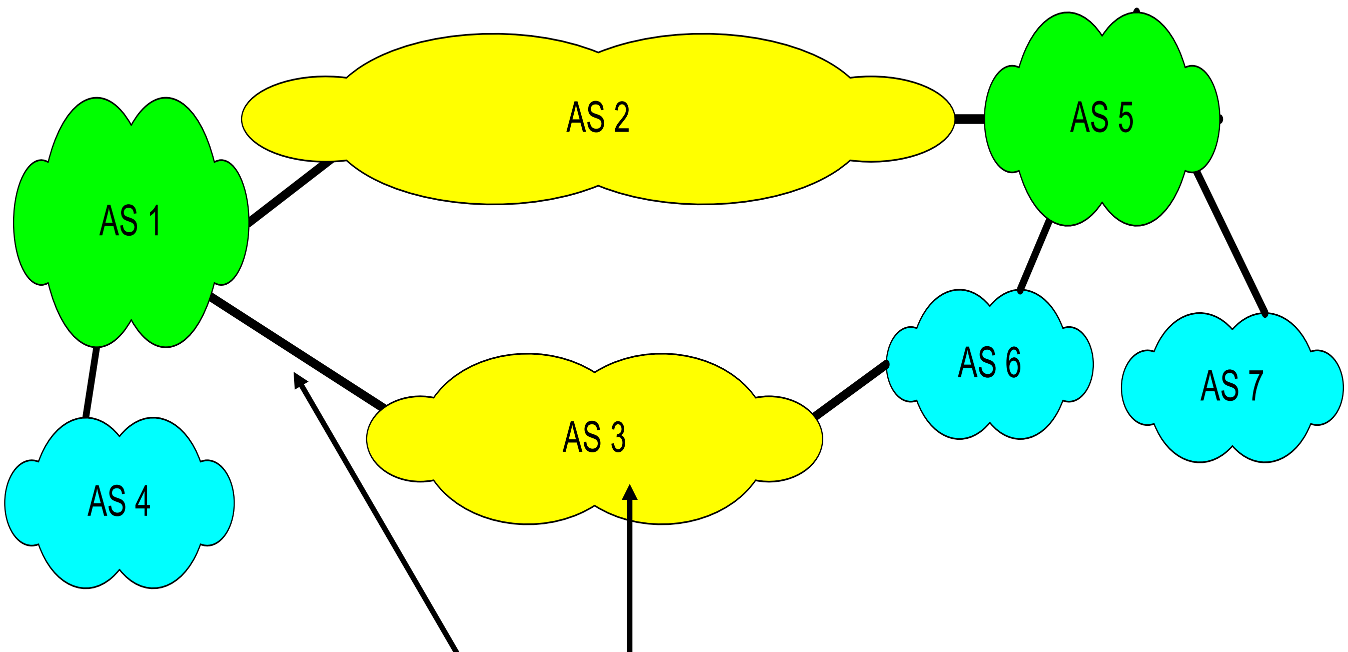
intra-AS routing

- routing among hosts, routers in same AS ("network")
- all routers in AS must run *same* intra-domain protocol
- routers in *different* AS can run *different* intra-domain routing protocol
- gateway router: at "edge" of its own AS, has link(s) to router(s) in other AS'es

inter-AS routing

- routing among AS'es
- gateways perform inter-domain routing (as well as intra-domain routing)

Interdomain and Intradomain Routing



- routing protocols used inside an AS, referred to as intradomain routing, are called interior gateway protocols (IGP)
  - objective: shortest path, only operate within an AS
- routing protocols used between ASs, referred to as interdomain routing, are called exterior gateway protocols (EGP)
  - objective: satisfy policy of the ASs, not always shortest path

Interdomain and Intradomain Routing

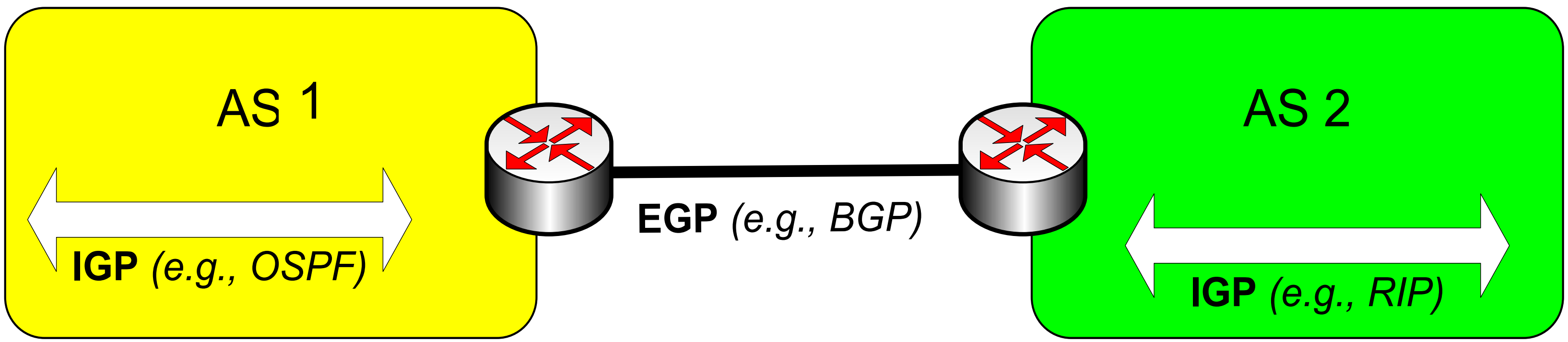
Intradomain Routing

- routing within an Autonomous System (AS)
- ignores the Internet outside the AS
- protocols for Intradomain routing are collectively called **Interior Gateway Protocols** or **IGP's**.
- popular protocols are:
  - RIP (simple, old)
  - OSPF (better)

Interdomain Routing

- routing between AS's
- assumes that the Internet consists of a collection of interconnected AS's
- normally, there is one dedicated router in each AS that handles interdomain traffic.
- protocols are collectively called **Exterior Gateway Protocols** or **EGP's.**
- popular protocols are:
  - Border Gateway Protocol (BGP) v4 current

EGP and IGP

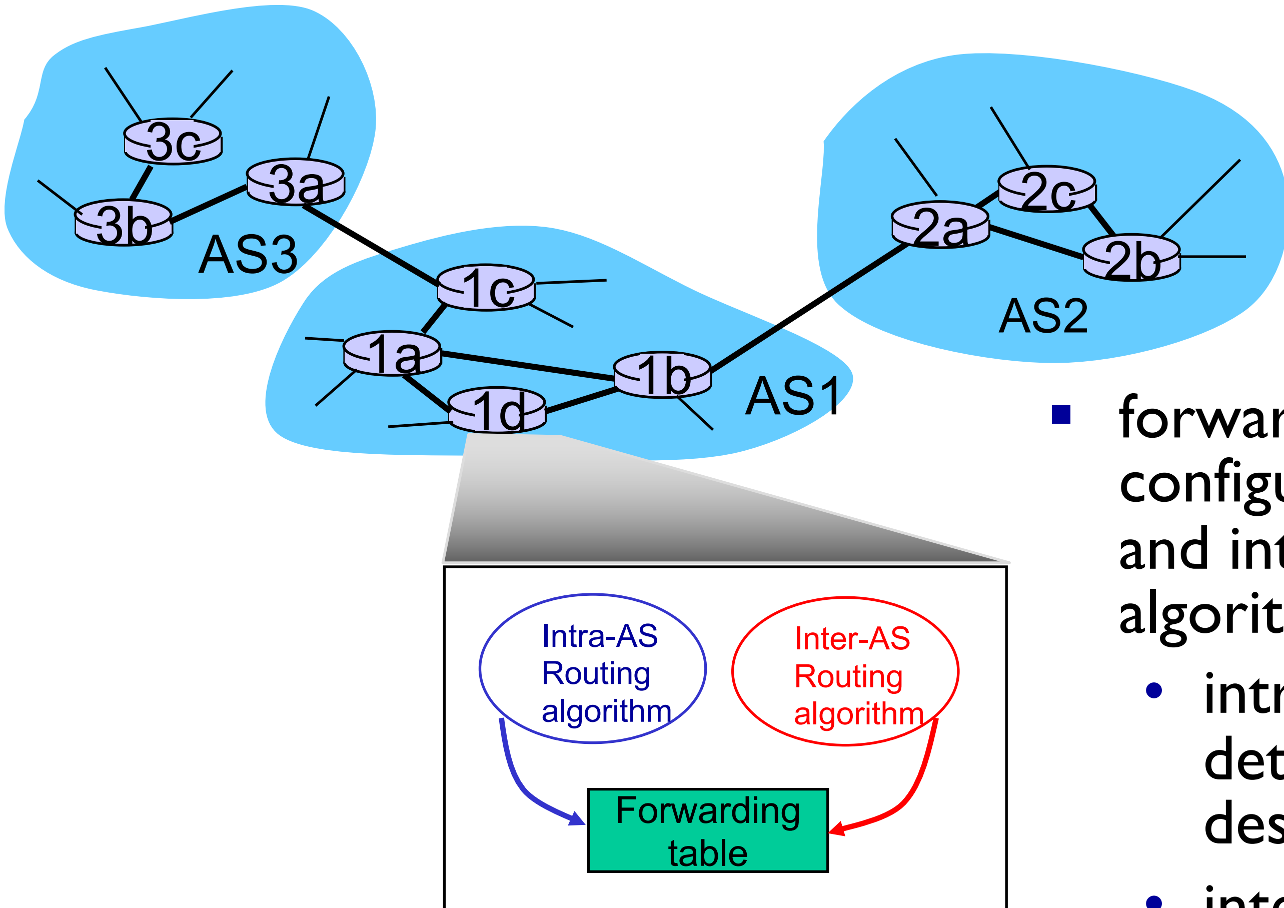


- Interior Gateway Protocol (IGP)
  - routing is done based on metrics
  - routing domain is one AS
- Exterior Gateway Protocol (EGP)
  - routing is done based on policies
  - routing domain is the entire Internet

Intra-AS Routing

- also known as *interior gateway protocols (IGP)*
- most common intra-AS routing protocols:
  - RIP: Routing Information Protocol
  - OSPF: Open Shortest Path First (IS-IS protocol essentially same as OSPF)
  - IGRP: Interior Gateway Routing Protocol (Cisco proprietary for decades, until 2016)

Interconnected ASes



- forwarding table configured by both intra- and inter-AS routing algorithm
  - intra-AS routing determine entries for destinations within AS
  - inter-AS & intra-AS determine entries for external destinations

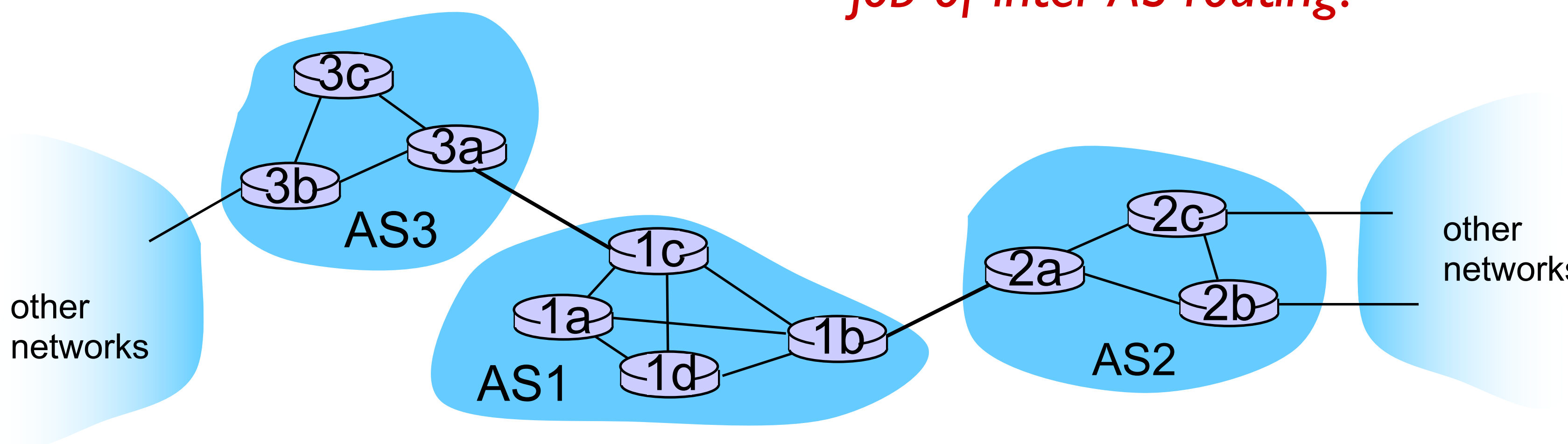
Inter-AS tasks

- suppose router in AS1 receives datagram destined outside of AS1:
  - router should forward packet to gateway router, but which one?

AS1 must:

1. learn which destinations are reachable through AS2, and which through AS3
2. propagate this reachability info to all routers in AS1

job of inter-AS routing!



Multiple Routing Protocols

- multiple routing protocols can run on the same router
- if a router is an exterior gateway router then usually one IGP and one EGP protocol will be in operation
- each routing protocol updates the routing table accordingly